Necessary and Sufficient Conditions for the Existence of an LU Factorization

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Abstract

We establish necessary and sufficient conditions for the existence of an LU factorization $A = LU$ for an arbitrary square matrix A, including singular and rank-deficient cases, without the use of row or column permutations. We prove that such a factorization exists if and only if the nullity of every leading principal submatrix is bounded by the sum of the nullities of the corresponding leading column and row blocks. While building upon the work of Okunev and Johnson, we present simpler, constructive proofs based on induction. Furthermore, we extend these results to characterize rank-revealing factorizations, providing explicit sparsity bounds for the factors L and U. Finally, we derive analogous necessary and sufficient conditions for the existence of factorizations constrained to have unit lower or unit upper triangular factors.

List of Corrections

It is well-known that for any square matrix A, there exists a permutation of rows (and potentially columns) such that an LU factorization exists. Specifically, the factorization $PA = LU$, where P is a permutation matrix, L is unit lower triangular, and U is upper triangular, is guaranteed to exist for any matrix A [2, 6].

However, the existence of an LU factorization for a matrix A *without* permutations—that is, $A = LU$—is not guaranteed. A classical result states that for a non-singular matrix, such a factorization exists if and only if all its leading principal minors are non-zero. When the matrix is singular or when leading principal minors vanish, the situation becomes significantly more complex. The standard conditions involving determinants are no longer sufficient to characterize the existence of the factorization.

This paper addresses the necessary and sufficient conditions for the existence of an LU factorization for an arbitrary matrix A, without imposing restrictions on its rank or invertibility. We extend the work of Okunev and Johnson [8], who established rank-based conditions for this problem (see also [1] for a discussion on multiple factorizations of singular matrices). Other perspectives on factorizations and their existence in general algebraic structures or specific contexts can be found in [7, 5, 4].

The contributions of this paper are threefold. First, we formulate the necessary and sufficient condition using the *nullity* of the leading principal submatrices relative to the nullity of the corresponding column and row blocks. Specifically, we prove that an LU factorization exists if and only if for all k:

$$\text{null}(A[1:k, 1:k]) \leq \text{null}(A[:, 1:k]) + \text{null}(A[1:k, :]^T)$$

Second, we provide new, simpler proofs of these results. Our proofs are constructive and rely on induction, explicitly building the factorization by handling the singular cases via internal permutations that preserve the triangular structure. Finally, we extend these results to characterize rank-revealing factorizations, providing tight sparsity bounds for the factors, and we derive analogous conditions for factorizations requiring unit lower or unit upper triangular factors.

The remainder of the paper is organized as follows. Section 1 reviews classical results regarding factorizations with permutations. Section 2 presents the main theorem and the derivation of the nullity condition. Theorem 2.3 strengthens this result by characterizing rank-revealing factorizations. Finally, Section 3 discusses the specific requirements for unit triangular factors.

We begin with a few classical results that will serve as reference for the later sections.

Notations

$\mathrm{rank}(A)$	Rank of matrix A
$\mathrm{null}(A)$	Dimension of the right null space of matrix A
A^T	Transpose of matrix A
a_{ij}	Entry (i,j) of matrix A
$A[:,i]$	Column i of matrix A
$A[i,:]$	Row i of matrix A
$A[i:j,k:l]$	Submatrix of A with rows from i to j and columns from k to l
$A[i:j,k:]$	Submatrix of A with rows from i to j and columns from k to the end
e_i	i-th vector of the standard basis
I_n	Identity matrix of size n

1 Classical results

We begin by recalling standard existence results involving permutations. These theorems correspond to Gaussian elimination with partial and full pivoting.

Theorem 1.1 (Partial Pivoting)**.** *For any square matrix A of size n, there exists a permutation matrix P such that $PA = LU$, where L is unit lower triangular and U is upper triangular.*

Proof. This is a classical result. We can prove it by induction on n. The result is true for $n = 1$.

Assume it is true for $n-1$. If $a_{11} \neq 0$, then consider:

$$A = LDU$$

where

$$L = \begin{pmatrix} 1 & 0 \\ a_{11}^{-1}A_{21} & I \end{pmatrix}, \quad D = \begin{pmatrix} 1 & 0 \\ 0 & S \end{pmatrix}, \quad U = \begin{pmatrix} a_{11} & A_{12} \\ 0 & I \end{pmatrix}$$

We can apply the induction hypothesis to S. The result for A follows.

If $a_{11} = 0$, we have two cases.

Case 1: column 1 of A is non-zero. We can find the smallest i such that $A[i,1] \neq 0$. Pivot row 1 and i. Denote by $B = \Pi_i A$ the permuted matrix (with $\Pi_i^2 = I$). Since $b_{11} \neq 0$, B has an LU factorization. This proves the result for A.

Case 2: column 1 of A is zero. We choose:

$$A = IDU$$

where

$$D = \begin{pmatrix} 1 & 0 \\ 0 & S \end{pmatrix}, \quad U = \begin{pmatrix} 0 & A_{12} \\ 0 & I \end{pmatrix}$$

and $S = A_{22}$. We can apply the induction hypothesis to S. The result for A follows. $\square$

Corollary 1.1. *For any square matrix A of size n, there exists a permutation matrix Q such that $AQ = LU$, where L is lower triangular and U is unit upper triangular.*

Proof. Apply Theorem 1.1 to A^T. □

Definition 1.1. *A matrix A of size $m \times n$ is said to have a* rank revealing LU factorization *if it can be written as $A = LU$, where L is a lower triangular matrix of size $m \times r$, U is an upper triangular matrix of size $r \times n$, and $r = \operatorname{rank}(A)$.*

If we allow both row and column permutations, we can always obtain such a factorization.

Theorem 1.2 (Full Pivoting)**.** *For any matrix A of size $m \times n$, there exist a row permutation matrix P and a column permutation matrix Q such that $PAQ = LU$ is a rank revealing LU factorization.*

Proof. This is also a classical result. We can prove it by induction on n. The result is true for $n = 1$.

Assume it is true for $n - 1$. If $a_{11} \neq 0$, then consider:

$$A = LDU$$

where

$$L = \begin{pmatrix} a_{11} & 0 \\ A_{21} & I \end{pmatrix}, \quad D = \begin{pmatrix} a_{11}^{-1} & 0 \\ 0 & S \end{pmatrix}, \quad U = \begin{pmatrix} a_{11} & A_{12} \\ 0 & I \end{pmatrix}$$

S has rank $r - 1$ if $r = \operatorname{rank}(A)$. We can apply the induction hypothesis to S. The result for A then follows.

If $a_{11} = 0$, we have two cases. If A is 0, we pick $L = U = 0$. Otherwise we have some i and j such that $a_{ij} \neq 0$. We can permute row 1 and i and column 1 and j. Denote by $B = \Pi_i A \Pi_j$ the permuted matrix. Since $b_{11} \neq 0$, B has a rank revealing LU factorization. This proves the result for A.

We have also proved that this process must terminate after r steps where $r = \operatorname{rank}(A)$. So the LU factorization of B is rank revealing.

From **??** and **??** (see Section A in the Appendix), we can also prove that the remaining Schur complement is zero after r steps of the LU factorization. □

We now consider LU factorizations without permutations. The fundamental result states that factorization is possible if and only if the matrix is *strongly non-singular* (i.e., all leading principal minors are non-zero).

Proposition 1.1. *Let A be a non-singular matrix of size n. A has an LU factorization if and only if all its leading principal submatrices are non-singular:*

$$\operatorname{rank}(A[1:k, 1:k]) = k, \quad k = 1, \ldots, n$$

Proof. We prove that the condition is necessary. Assume $A = LU$:

$$\det(A) = \prod_{i=1}^{n} l_{ii} u_{ii}$$

and $\det(A) \neq 0$. So, $l_{ii} \neq 0$ and $u_{ii} \neq 0$ for all i. Therefore: $\operatorname{rank}(A[1:k, 1:k]) = k$, $k = 1, \ldots, n$.

We prove that the condition is sufficient. We prove the result by induction on n. The result is true for $n = 1$.

Assume it is true for $n - 1$. By assumption, $a_{11} \neq 0$. We then have the factorization:

$$A = LDU$$

where

$$L = \begin{pmatrix} a_{11} & 0 \\ A_{21} & I \end{pmatrix}, \quad D = \begin{pmatrix} a_{11}^{-1} & 0 \\ 0 & S \end{pmatrix}, \quad U = \begin{pmatrix} a_{11} & A_{12} \\ 0 & I \end{pmatrix}$$

and $S = A_{22} - A_{21} a_{11}^{-1} A_{12}$ is the Schur complement. L and U are full rank. So D satisfies the condition of the theorem. Consequently, S also satisfies the condition of the theorem and is of size $n - 1$. By the induction hypothesis, S has an LU factorization. Therefore A also has an LU factorization. □

2 General necessary and sufficient condition for the existence of an LU factorization

We now address the central question of this paper: determining the necessary and sufficient conditions for the existence of an LU factorization without permutations. We begin with two illustrative examples that motivate the general result.

Consider the matrix:

$$A = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

This matrix does not admit an LU factorization. We prove this by contradiction. Suppose $A = LU$. Then the diagonal entry satisfies $a_{11} = l_{11}u_{11} = 0$. This implies that either $l_{11} = 0$ or $u_{11} = 0$.

- If $l_{11} = 0$, then $a_{12} = l_{11}u_{12} = 0$, which contradicts $a_{12} = 1$.
- If $u_{11} = 0$, then $a_{21} = l_{21}u_{11} = 0$, which contradicts $a_{21} = 1$.

In this special case, we have $A^2 = I$ and A is a permutation matrix. So we note that $AA = I \cdot I$ is the factorization of Theorem 1.1 with $P = A$, $L = I$, and $U = I$.

However, a zero pivot does not always preclude factorization. Consider the singular matrix:

$$A = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

Despite the zero principal minors, this matrix *does* have an LU factorization. We can derive it by permuting A into a factorizable form and then reversing the permutations. Apply a cyclic row shift $(1, 2, 3) \to (2, 3, 1)$ via P and swap columns 1 and 3 via Q:

$$PAQ = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}$$

We can recover a factorization for A by writing $A = P^{-1}(PAQ)Q^{-1}$:

$$A = P^{-1} \left[\begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} \right] Q^{-1} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

This yields a valid, rank-revealing LU factorization with no pivoting. Another valid LU factorization is:

$$A = \begin{pmatrix} 0 & 0 \\ 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

We can generalize this result using block matrices. Let 0_n denote the $n \times n$ zero matrix. If B and C are factorizable matrices such that $B = LU$ and $C = LU$, we can construct larger factorizable matrices A as follows:

$$A = \begin{pmatrix} 0_n & 0_n & 0_n \\ 0_n & 0_n & C \\ 0_n & B & D \end{pmatrix} = \begin{pmatrix} 0_n & 0_n \\ C & 0_n \\ D & L \end{pmatrix} \begin{pmatrix} 0_n & 0_n & I_n \\ 0_n & U & 0_n \end{pmatrix}$$

Alternatively, using the factorization of C:

$$A = \begin{pmatrix} 0_n & 0_n \\ 0_n & L \\ I_n & 0_n \end{pmatrix} \begin{pmatrix} 0_n & B & D \\ 0_n & 0_n & U \end{pmatrix}$$

These examples demonstrate that the existence of a factorization is linked to the relationship between the ranks (or nullities) of specific sub-blocks. We formalize this observation in Property 2.1. We will show that Property 2.1 below is a necessary and sufficient condition for the existence of an LU factorization.

Property 2.1. *Matrix A is square of size n. For all $k = 1, \ldots, n$:*

$$\operatorname{null}(A[1:k, 1:k]) \leq \operatorname{null}(A[:, 1:k]) + \operatorname{null}(A[1:k, :]^T) \tag{1}$$

We informally describe the main strategy before stating the results formally. The primary obstacle to constructing an LU factorization is the occurrence of a zero pivot. Standard Gaussian elimination resolves this by permuting rows (partial pivoting) or both rows and columns (full pivoting) to bring a non-zero entry to the diagonal.

However, arbitrary permutations destroy the triangular structure required for the specific factorization $A = LU$. If we write the permuted factorization as $PAQ = LU$, recovering A yields $A = (P^{-1}L)(UQ^{-1})$. In general, $P^{-1}L$ is not lower triangular and UQ^{-1} is not upper triangular.

To maintain the triangular factors of A, we must restrict our pivoting strategy. We rely on the observation (detailed in Section A) that a zero column (or row) in the Schur complement corresponds to a column (or row) in the original matrix that is linearly dependent on the preceding columns (or rows).

Our constructive proof effectively "sorts" the matrix indices. Whenever we encounter a zero pivot, the condition in Property 2.1 guarantees that at least one of the following two scenarios must hold:

1. The current column in the Schur complement is zero (the column is linearly dependent). We permute this column to rightward.

2. The current row in the Schur complement is zero (the row is linearly dependent). We permute this row downward.

Crucially, because we only permute *zero* columns or rows (relative to the active Schur complement), the triangular structure of the resulting factors is preserved. When we reverse these permutations, the factor $P^{-1}L$ remains lower triangular and UQ^{-1} remains upper triangular.

Condition (1) is the precise safeguard that ensures we never reach a "dead end" where the pivot is zero but neither the column nor the row can be safely permuted away (i.e., a case where the rank has not been exhausted, but the leading block is rank-deficient in a way that blocks factorization).

Consequently, this process produces a rank-revealing factorization where the indices are partitioned based on linear independence:

- **Linearly Independent Vectors:** If a column j_0 of A is linearly independent of the previous columns, it is retained (or moved) into the leading rank-r block ($j \leq r$).

- **Linearly Dependent Vectors:** If a column j_0 of A is linearly dependent on the previous columns, it is deferred to the trailing block ($j > r$).

A symmetric logic applies to the rows of A.

We now provide the formal proof of the main results, Theorems 2.2 and 2.3.

2.1 Rank theorems

We recall rank theorems that will be useful in the proof of Theorem 2.2.

Theorem 2.1. *The rank-nullity theorem states that for any matrix A of size $m \times n$:*

$$\operatorname{rank}(A) + \operatorname{null}(A) = n$$

We now state Sylvester's inequality for the rank of a product of matrices.

Lemma 2.1. *Consider $A = BC$. Then:*

$$\operatorname{rank}(A) \leq \min(\operatorname{rank}(B), \operatorname{rank}(C)), \quad \operatorname{rank}(B) + \operatorname{rank}(C) - k \leq \operatorname{rank}(A)$$

where k is the number of columns of B and rows of C.

Corollary 2.1. *If A, B, and C are square matrices then:*

$$\operatorname{null}(A) \leq \operatorname{null}(B) + \operatorname{null}(C)$$

Proof. Assume $A = BC$ where A, B, C are $n \times n$ matrices. From Lemma 2.1:

$$\text{rank}(A) \geq \text{rank}(B) + \text{rank}(C) - n$$

Using Theorem 2.1, $\text{rank}(M) = n - \text{null}(M)$:

$$n - \text{null}(A) \geq (n - \text{null}(B)) + (n - \text{null}(C)) - n$$

Simplifying: $\text{null}(A) \leq \text{null}(B) + \text{null}(C)$. □

2.2 Necessary condition

We show that Property 2.1 is a necessary condition for the existence of an LU factorization.

Assume that A, a square matrix of size n, has an LU factorization:

$$A = LU$$

Then:

$$\begin{aligned} A[1:k,:] &= L[1:k,1:k]U[1:k,:], \\ A[:,1:k] &= L[:,1:k]U[1:k,1:k], \\ A[1:k,1:k] &= L[1:k,1:k]U[1:k,1:k] \end{aligned}$$

Using Lemma 2.1 and Corollary 2.1, we have:

$$\begin{aligned} \text{rank}(A[1:k,:]) &\leq \text{rank}(L[1:k,1:k]), \\ \text{rank}(A[:,1:k]) &\leq \text{rank}(U[1:k,1:k]), \\ \text{null}(A[1:k,1:k]) &\leq \text{null}(L[1:k,1:k]) + \text{null}(U[1:k,1:k]) \end{aligned}$$

Using Theorem 2.1, we get:

$$\text{null}(A[1:k,1:k]) \leq \text{null}(A[:,1:k]) + \text{null}(A[1:k,:]^T)$$

So Property 2.1 is a necessary condition for the existence of an LU factorization.

2.3 Sufficient condition

We will now prove the main theorem:

Theorem 2.2. *Matrix A satisfies Property 2.1 if and only if it has an LU factorization.*

Proof. We prove the result by induction on the size of A. The result is true for $n = 1$.

Assume it is true for size $n - 1$.

Assume $a_{11} \neq 0$. We have the factorization:

$$A = LDU$$

where

$$L = \begin{pmatrix} a_{11} & 0 \\ A_{21} & I \end{pmatrix}, \quad D = \begin{pmatrix} a_{11}^{-1} & 0 \\ 0 & S \end{pmatrix}, \quad U = \begin{pmatrix} a_{11} & A_{12} \\ 0 & I \end{pmatrix}$$

Since L and U are triangular and full rank, D satisfies Property 2.1 if and only if A does. Consequently, S also satisfies Property 2.1 and is of size $n - 1$. Moreover S has rank $r - 1$ if $r = \text{rank}(A)$. By the induction hypothesis, S has an LU factorization. The result for A follows.

Assume $a_{11} = 0$. From Property 2.1, we have either:

1. $A[:,1] = 0$, $A[1,:] \neq 0$, or
2. $A[1,:] = 0$, $A[:,1] \neq 0$, or

3. $A[:,1] = 0$, $A[1,:] = 0$.

Case 1: $A[:,1] = 0$. We can find the smallest j such that $A[1,j] \neq 0$. Pivot column 1 and j. Denote by $B = A\Pi_j$ the permuted matrix (with $\Pi_j^2 = I$). We prove that B satisfies Property 2.1.

Consider $k < j$. We have

$$\operatorname{null}(A[1:k,1:k]) \leq \operatorname{null}(A[:,1:k]) + \operatorname{null}(A[1:k,:]^T) \tag{2}$$

Since column 1 of A is zero, we have:

$$\operatorname{null}(A[1:k,1:k]) = \operatorname{null}(A[1:k,2:k]) + 1$$

We have

$$\operatorname{null}(B[1:k,2:k]) = \operatorname{null}(A[1:k,2:k])$$

Since the only non-zero entry in row 1 of $B[1:k,1:k]$ is in column 1, $B[1:k,1]$ is linearly independent of $B[1:k,2:k]$. Therefore, we have

$$\operatorname{null}(B[1:k,1:k]) = \operatorname{null}(B[1:k,2:k]) = \operatorname{null}(A[1:k,2:k]) = \operatorname{null}(A[1:k,1:k]) - 1$$

By a similar reasoning, we have

$$\operatorname{null}(B[:,1:k]) = \operatorname{null}(A[:,1:k]) - 1$$

The column pivoting does not affect the rank of the rows, so

$$\operatorname{null}(B[1:k,:]^T) = \operatorname{null}(A[1:k,:]^T)$$

Therefore B satisfies Property 2.1 for $k < j$.

For $k \geq j$, the pivoting does not affect the ranks in Equation (2), so B also satisfies Property 2.1 for $k \geq j$.

So B satisfies Property 2.1.

Since $b_{11} \neq 0$, B has an LU factorization: $B = LU$. Column j of B is zero. If column j of U is non-zero, we can zero it out without affecting any column in LU. So we can assume that column j of U is zero. We have:

$$A = B\Pi_j = LU\Pi_j$$

and $U\Pi_j$ is still upper triangular because Π_j only permutes column 1 and j. Therefore A has an LU factorization.

Case 2: $A[1,:] = 0$. Consider $B = A^T$. B corresponds to Case 1, so B has an LU factorization: $B = LU$. Therefore:

$$A = B^T = U^T L^T$$

is an LU factorization of A.

Case 3: If A is 0, it has an LU factorization with $L = U = 0$.

Otherwise, if $A \neq 0$, let's denote by j the smallest index such that column j is non zero. Let's denote by $i > 1$ the smallest index such that $a_{ij} \neq 0$. Let's pivot column 1 and j and denote by $B = A\Pi_j$ the permuted matrix. B satisfies Case 2, so it has an LU factorization: $B = LU$. Column j of U is zero. Therefore:

$$A = B\Pi_j = LU\Pi_j$$

is an LU factorization of A.

We have also proved that this process must terminate after r steps where $r = \operatorname{rank}(A)$. So the LU factorization of B is rank revealing. □

We now prove a stronger theorem. Please compare this result with Theorem 1.2.

Theorem 2.3. *Let A be a matrix of size n with rank $r \leq n$. If A satisfies Property 2.1, then there exist a row permutation P_r and a column permutation P_c such that the matrix*

$$B = P_r^{-1} A P_c^{-1}$$

has strongly non-singular leading principal submatrices $B[1:k, 1:k]$ for $1 \leq k \leq r$. Consequently, the leading principal block admits an LU factorization $B[1:r, 1:r] = L_r U_r$.

Define the rank-revealing factors of B as:

$$L_B = \begin{pmatrix} L_r \\ B[r+1:n, 1:r]U_r^{-1} \end{pmatrix} \quad \text{and} \quad U_B = \begin{pmatrix} U_r & L_r^{-1}B[1:r, r+1:n] \end{pmatrix}$$

Then $B = L_B U_B$. Furthermore, A admits a rank-revealing factorization

$$A = LU$$

where $L = P_r L_B$ and $U = U_B P_c$.

Moreover, we have the following sparsity results for L. Let i_0 be the row index in A corresponding to the logical row index i in B (i.e., $e_{i_0} = P_r e_i$). Then:

- *If $i \leq r$, then $i_0 \geq i$ and $L[i_0, i+1:] = 0$.*
- *If $i > r$, then $i_0 \leq i$; if $i_0 \leq r$, then $L[i_0, i_0:] = 0$.*

Similarly for U, let j_0 be the column index in A corresponding to the logical column index j in B (i.e., $e_{j_0} = P_c^T e_j$). Then:

- *If $j \leq r$, then $j_0 \geq j$ and $U[j+1:, j_0] = 0$.*
- *If $j > r$, then $j_0 \leq j$; if $j_0 \leq r$, then $U[j_0:, j_0] = 0$.*

Proof. We follow a proof similar to the one for Theorem 2.2.

We do an induction on the size of A. The result is true for $n = 1$.

Assume it is true for size $n-1$. If $a_{11} \neq 0$, we have the factorization:

$$A = LDU$$

where

$$L = \begin{pmatrix} a_{11} & 0 \\ A_{21} & I \end{pmatrix}, \quad D = \begin{pmatrix} a_{11}^{-1} & 0 \\ 0 & S \end{pmatrix}, \quad U = \begin{pmatrix} a_{11} & A_{12} \\ 0 & I \end{pmatrix}$$

We apply the induction hypothesis to S. We can define P_r and P_c in terms of the permutations for S:

$$P_r = \begin{pmatrix} 1 & 0 \\ 0 & P_{r,S} \end{pmatrix}, \quad P_c = \begin{pmatrix} 1 & 0 \\ 0 & P_{c,S} \end{pmatrix}$$

The sparsity results follow directly from the induction hypothesis.

Assume $a_{11} = 0$. From Property 2.1, we have:

1. $A[:, 1] = 0$, $A[1, :] \neq 0$, or
2. $A[1, :] = 0$, $A[:, 1] \neq 0$, or
3. $A[:, 1] = 0$, $A[1, :] = 0$.

Case 1: $A[:, 1] = 0$. As for Theorem 2.2, we can find the smallest j such that $A[1, j] \neq 0$. We pivot column 1 and j. Denote by $B = A\Pi_j$ the permuted matrix. B satisfies Property 2.1 and $b_{11} \neq 0$. So B has an LU factorization with the sparsity pattern of Theorem 2.3. Since column j of B is zero, column j of U_B is zero. We define

$$P_c = P_{c,B}\Pi_j, \quad U = U_B \Pi_j$$

Π_j swaps columns 1 and j of U_B. We have $e_1 = \Pi_j^T e_j$ and column j of U_B is zero. We also have a j_0 such that $e_j = P_{c,B}^T e_{j_0}$ and $e_1 = P_c^T e_{j_0}$. Note that since column j of B (which is column 1 of A) is zero, we are beyond the rank-revealing part and $j_0 > r$. Since column 1 of U is zero and the other columns of U have the sparsity pattern of Theorem 2.3, U also satisfies the sparsity pattern of Theorem 2.3.

Case 2: $A[1,:] = 0$. Consider $B = A^T$. B corresponds to Case 1 and satisfies the sparsity pattern of Theorem 2.3. So A also satisfies the sparsity pattern of Theorem 2.3.

Case 3: If $A = 0$, it has an LU factorization. In that case we pick $L = U = 0$ as the factors. If $A \neq 0$, let's denote by j the smallest index such that column j is non zero. Let's denote by $i > 1$ the smallest index such that $a_{ij} \neq 0$. Let's pivot column 1 and j and denote by $B = A\Pi_j$ the permuted matrix. B satisfies Case 2, so it has an LU factorization: $B = LU$ with the sparsity pattern of Theorem 2.3. Following a reasoning analogous to Case 1, we can show that A also satisfies the sparsity pattern of Theorem 2.3.

As in Theorem 2.2, we have also proved that this process must terminate after r steps where $r = \mathrm{rank}(A)$. So the LU factorization of B is rank revealing. □

FiXme Note!

FiXme Note: Add a Python code of the algorithm to compute the LU factorization with the required pivoting strategy. The algorithm only fails if the matrix does not satisfy Property 2.1.

3 Unit triangular factors

FiXme Note!

FiXme Note: Explain the difference with the previous row pivoting strategy to get a unit lower triangular factor; the previous result requires a row pivoting in order to get a unit lower triangular factor; here we require a column pivoting; this column pivoting only works under restricted conditions. For a general matrix, it's not possible to do column pivoting only to get a unit lower triangular factor. Only row pivoting allows that. Provide counterexamples to illustrate this. The main difference of course is that we get an LU factorization while the classical result only provides an LUP factorization.

We now examine the specific case where one of the factors is required to be unit triangular. The following result should be compared with Theorem 1.1.

Theorem 3.1. *Let A be a matrix of size n and rank $r \leq n$. A has an LU factorization $A = LU$ where L is square unit lower triangular if and only if for all $k = 1, \ldots, n$:*

$$\mathrm{null}(A[1:k, 1:k]) = \mathrm{null}(A[:, 1:k]) \tag{3}$$

Construction: *If A satisfies Equation (3), there exists a column permutation P_c such that the matrix $B = AP_c^{-1}$ admits a factorization $B = L_B U_B$ where L_B is unit lower triangular. Consequently,*

$$A = L_B(U_B P_c) = LU$$

is a valid factorization of A with $L = L_B$ being unit lower triangular.
Structure: *For the matrix B constructed above, row j is a linear combination of the previous rows if and only if column j is a linear combination of the previous columns.*
Sparsity: *Let j_0 be the column index in A corresponding to the logical column index j in B (i.e., $e_{j_0} = P_c^T e_j$). The factors exhibit the following sparsity patterns:*

- ***Dependent Case:*** *If row j of B is a linear combination of the previous rows, then $j_0 \leq j$. Furthermore:*

$$L[:, j] = e_j, \quad U[j,:] = 0, \quad \textit{and} \quad U[j_0:, j_0] = 0$$

- ***Independent Case:*** *If row j of B is linearly independent of the previous rows, then $j_0 \geq j$. Furthermore:*

$$U[j+1:, j_0] = 0$$

Proof. We first prove that if $A = LU$ with L unit lower triangular, then A satisfies Equation (3). U trivially satisfies Equation (3) since it is upper triangular. Since $L[1:k, 1:k]$ is full rank, A also satisfies Equation (3).

We follow a proof similar to the one for Theorem 2.3. We do an induction on the size of A. The result is true for $n = 1$.

Assume it is true for size $n-1$. If $a_{11} \neq 0$, we verify that the result is true by the induction hypothesis.

Assume $a_{11} = 0$. From Equation (3), we have $A[:, 1] = 0$.

Case 1: If row 1 of A is non-zero, we can use the same steps as in the proof as Theorem 2.3 with a column permutation to prove the result by induction. In addition, this shows that if $e_{j_0} = P_c^T e_j$ and if row j of B is linearly independent of the previous rows then $j_0 \geq j$ and $U[j+1 :, j_0] = 0$.

Case 2: If row 1 of A is zero, we cannot use the same proof as Theorem 2.3 because we can no longer permute rows. However, since row 1 of A is zero, we can define:

$$A = IDU$$

with

$$D = \begin{pmatrix} 1 & 0 \\ 0 & S \end{pmatrix}, \quad U = \begin{pmatrix} 0 & A[1, 2:n] \\ 0 & I \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & I \end{pmatrix}$$

and $S = A_{22}$. D satisfies

$$\text{null}(D[1:k, 1:k]) = \text{null}(A[1:k, 1:k]) - 1 = \text{null}(A[:, 1:k]) - 1 = \text{null}(D[:, 1:k])$$

D satisfies Equation (3). By the induction hypothesis, we can find a column permutation $P_{c,S}$ such that $SP_{c,S}^{-1}$ has an LU factorization with unit lower triangular factor.

This also proves that row j of B is a linear combination of the previous rows if and only if column j of B is a linear combination of the previous columns. Moreover, if $e_{j_0} = P_c^T e_j$ and row j of B is a linear combination of the previous rows, then $j_0 \leq j$. We have $L[:, j] = e_j$, $U[j, :] = 0$, and $U[j_0 :, j_0] = 0$.

Note that the step above with D is not a valid step in Theorem 2.3. Indeed:

$$\begin{aligned} \text{null}(D[1:k, 1:k]) &= \text{null}(A[1:k, 1:k]) - 1 \\ \text{null}(D[:, 1:k]) &= \text{null}(A[:, 1:k]) - 1 \\ \text{null}(D[1:k, :]^T) &= \text{null}(A[1:k, :]^T) - 1 \end{aligned}$$

So that Property 2.1 may not be satisfied by D. □

We now state the corresponding result where U is unit upper triangular. Please compare the result below with Corollary 1.1.

Corollary 3.1. *Let A be a matrix of size n and rank $r \leq n$. A has an LU factorization $A = LU$ where U is unit upper triangular if and only if for all $k = 1, \ldots, n$:*

$$\text{null}(A[1:k, 1:k]) = \text{null}(A[1:k, :]) \tag{4}$$

Construction: *If A satisfies Equation (4), there exists a row permutation P_r such that the matrix $B = P_r^{-1} A$ admits a factorization $B = L_B U_B$ where U_B is unit upper triangular. Consequently,*

$$A = (P_r L_B) U_B = LU$$

is a valid factorization of A with $U = U_B$ being unit upper triangular.
Structure: *For the matrix B constructed above, column i is a linear combination of the previous columns if and only if row i is a linear combination of the previous rows.*
Sparsity: *Let i_0 be the row index in A corresponding to the logical row index i in B (i.e., $e_{i_0} = P_r e_i$). The factors exhibit the following sparsity patterns:*

- ***Dependent Case:*** *If column i of B is a linear combination of the previous columns, then $i_0 \leq i$. Furthermore:*
$$U[i, :] = e_i, \quad L[:, i] = 0, \quad \text{and} \quad L[i_0, i_0 :] = 0$$
- ***Independent Case:*** *If column i of B is linearly independent of the previous columns, then $i_0 \geq i$. Furthermore:*
$$L[i_0, i+1 :] = 0$$

Proof. Consider $B = A^T$. Apply Theorem 3.1. □

4 Conclusion

In this paper, we have provided an analysis of the necessary and sufficient conditions for the existence of an LU factorization for arbitrary square matrices, without the use of pivoting. We established that the existence of such a factorization is determined by a specific inequality relating the nullity of the leading principal submatrices to the nullity of the corresponding leading column and row blocks. This condition generalizes the classical requirement for non-singular matrices (non-vanishing leading principal minors) to the general rank-deficient case.

Our approach utilizes a constructive proof based on induction. This method not only simplifies the theoretical derivation but also provides explicit insight into how singular blocks can be handled through internal permutations that preserve the overall triangular structure of the factors.

Furthermore, we defined and characterized rank-revealing LU factorizations without global pivoting. We derived tight sparsity bounds for the resulting factors L and U, linking the physical indices of the non-zero entries to the logical rank profile of the matrix. Finally, we examined the constrained cases where one factor is required to be unit triangular, identifying the specific rank conditions required and their duality with respect to row and column permutations. These results offer a unified framework for understanding matrix factorizations in the presence of singularity and rank deficiency.

A Appendix

We prove a few additional results that are helpful in the main proofs. For additional background on connections between the LU factorization, Schur complements, and oblique projections, please refer to [9, 10, 3].

Lemma A.1. *Denote by X and Y two matrices of size $n \times k$. Assume that X is full rank and that $Y^T X$ is non-singular. Define:*

$$P = X(Y^T X)^{-1} Y^T$$

Then P is the oblique projection onto the range of X along the orthogonal complement of the range of Y.

Proof. We have:

$$PX = X(Y^T X)^{-1} Y^T X = X$$

So the range of P contains the range of X. Since $\text{rank}(P) \leq k$ and $\text{rank}(X) = k$, the range of P is exactly the range of X.

Consider v in the orthogonal complement of the range of Y (i.e., $Y^T v = 0$). We have:

$$Pv = X(Y^T X)^{-1} Y^T v = 0$$

Thus, the null space of P contains the orthogonal complement of the range of Y. By the rank-nullity theorem, the dimensions match, so the null space of P is exactly the orthogonal complement of the range of Y. □

Lemma A.2. *Denote by $X = [x_1, \ldots, x_k]$ a rank k matrix. Consider the oblique projection $P_o(X)$ onto*

$$S = \text{span}\{x_1, \ldots, x_k\}$$

along the subspace of vectors with the first k entries equal to zero. Assume that $X[1:k,:]$ is non-singular. Then:

$$P_o(X) = X(X[1:k,:])^{-1} Y^T$$

where

$$Y = \begin{pmatrix} I_k \\ 0 \end{pmatrix}$$

Proof. In Lemma A.1, set Y as

$$Y = \begin{pmatrix} I_k \\ 0 \end{pmatrix}$$

The range of Y is the span of the first k standard basis vectors. Its orthogonal complement is exactly the subspace of vectors with the first k entries equal to zero. □

Denote by $Q_o(X) = I - P_o(X)$ the oblique projection onto the subspace of vectors with the first k entries equal to zero along $S = \text{span}(X)$.

Consider the state of the matrix at the beginning of step k of the LU factorization algorithm (after $k-1$ columns have been eliminated). The Schur complement can be interpreted in terms of the oblique projection Q_o. Specifically, from Lemma A.2 applied to the first $k-1$ columns, we have:

$$S_k^o = A[:, k:] - A[:, 1:k-1]A[1:k-1, 1:k-1]^{-1}A[1:k-1, k:] = Q_o(A[:, 1:k-1])\ A[:, k:]$$

The standard Schur complement S_k corresponds to the bottom block of this projected matrix: $S_k = S_k^o[k:, :]$.

Lemma A.3. *If column j of A (where $j \geq k$) is linearly dependent on columns 1 to $k-1$ of A, then column $j-k+1$ of S_k^o is zero.*

Proof. Since column j of A is linearly dependent on columns 1 to $k-1$, we have:

$$A[:, j] \in \text{span}\{A[:, 1], \ldots, A[:, k-1]\}$$

Let $X = A[:, 1:k-1]$. Then $A[:, j] \in \text{span}(X)$. By definition of the projection $P_o(X)$:

$$P_o(X)A[:, j] = A[:, j]$$

Therefore:

$$Q_o(X)A[:, j] = (I - P_o(X))A[:, j] = 0$$

Since S_k^o is formed by applying $Q_o(X)$ to the columns $A[:, k:]$, the column corresponding to $A[:, j]$ is zero. $\square$

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